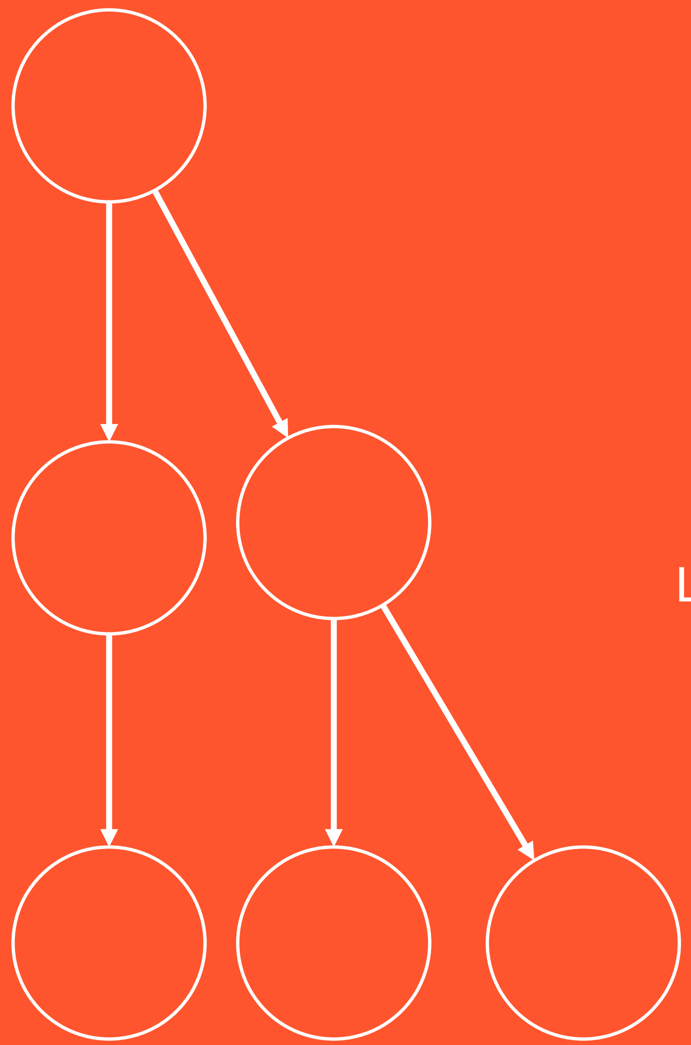


BCOG 100

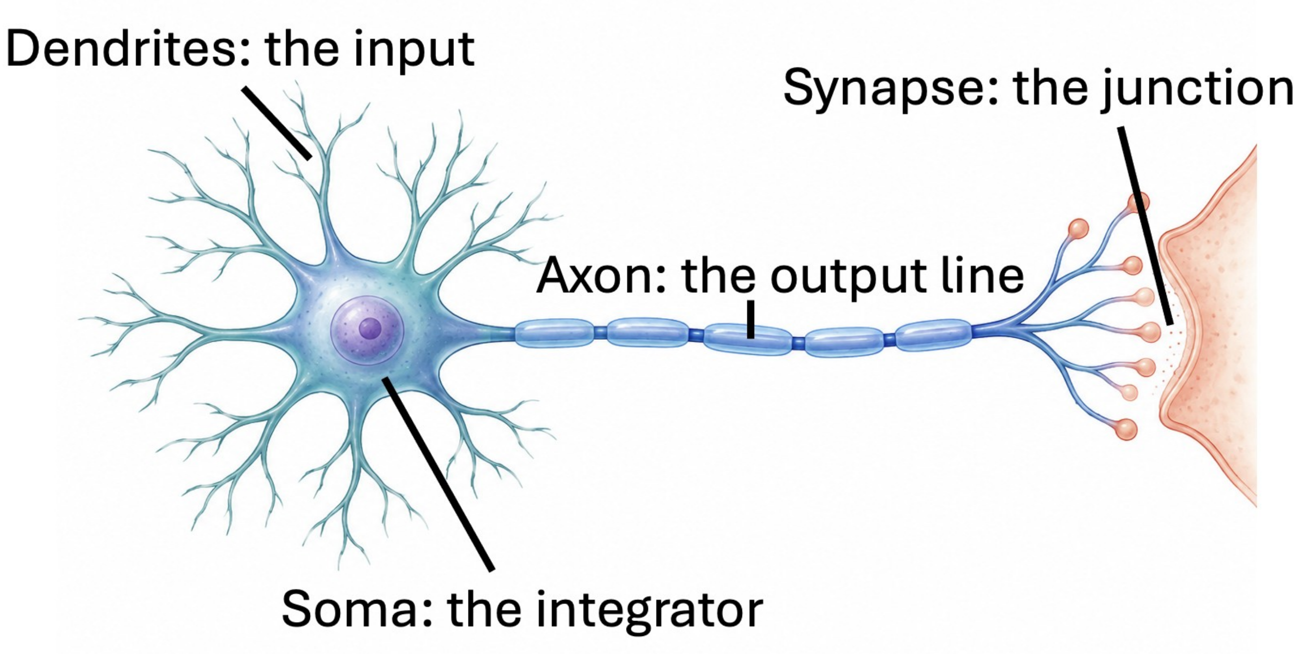
Neurons and Neural Communication

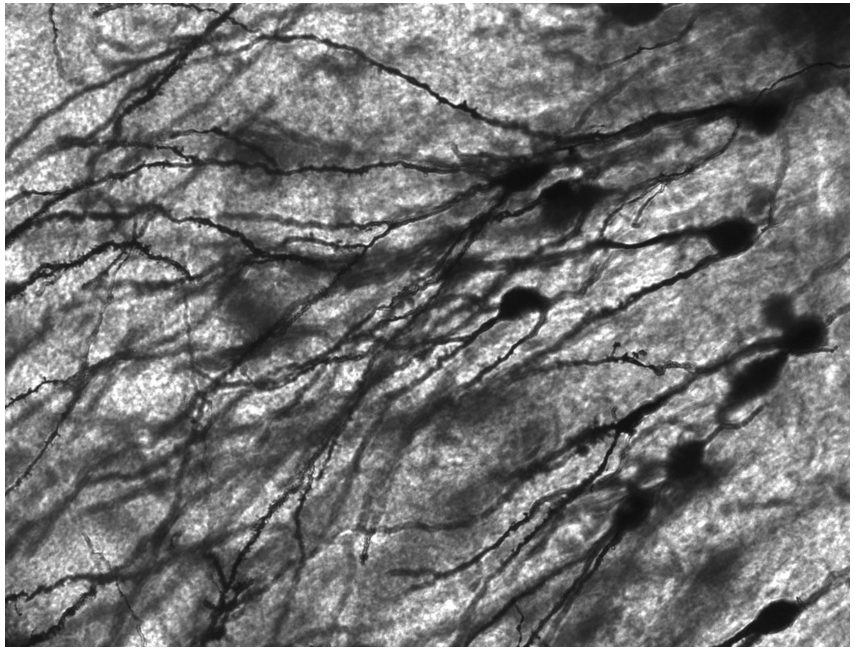
Lecture 3.3 — The Biology of Neurons, including Metabolism

UNIVERSITY OF ILLINOIS
URBANA-CHAMPAIGN

Four Roles, and the Parts That Fill Them

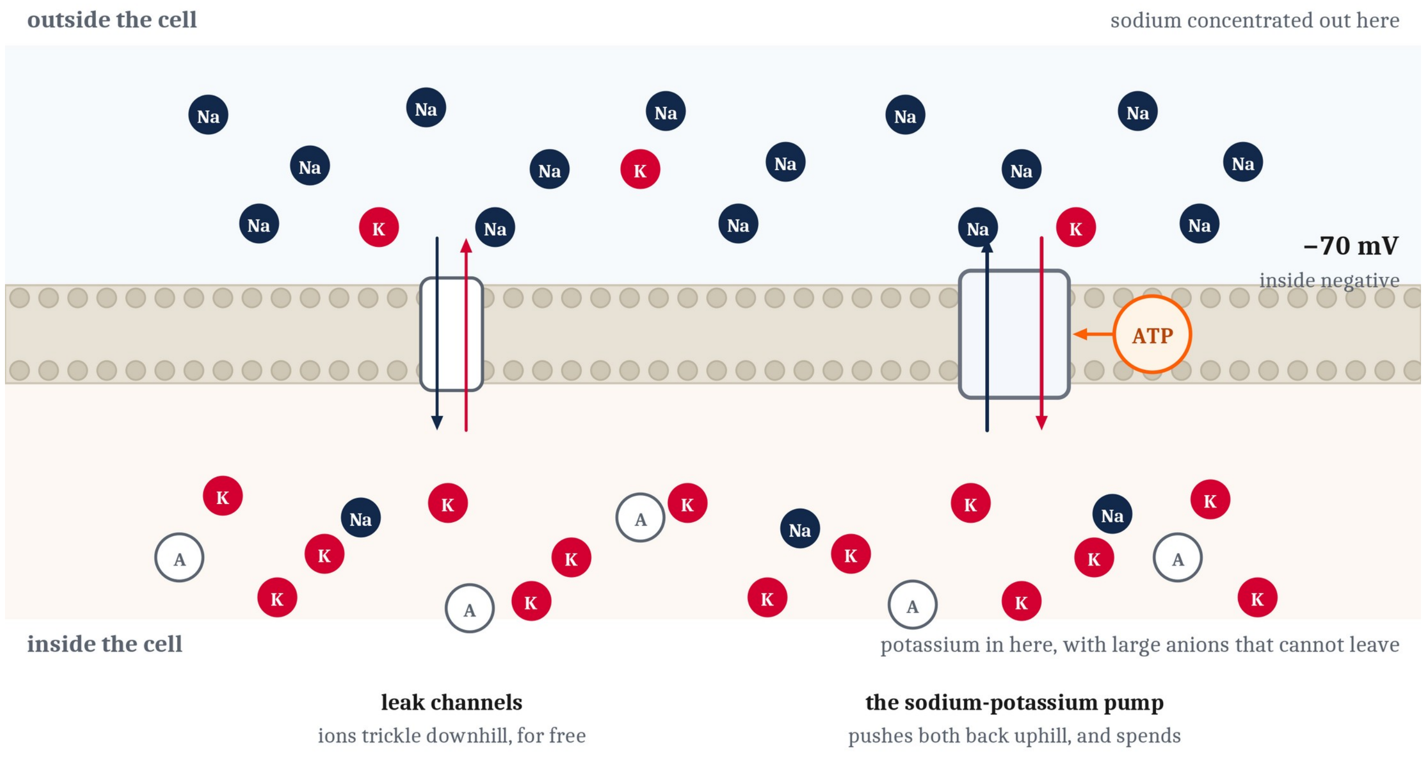
The input surface is the **dendrites**. The integrator is the **soma**, the cell body. The output line is the **axon**, a single fiber that may run a meter in a large animal. The junction is the **synapse**, the narrow gap between an axon's ending and the next cell's dendrites.





A Neuron at Rest Is Not a Neuron Switched Off

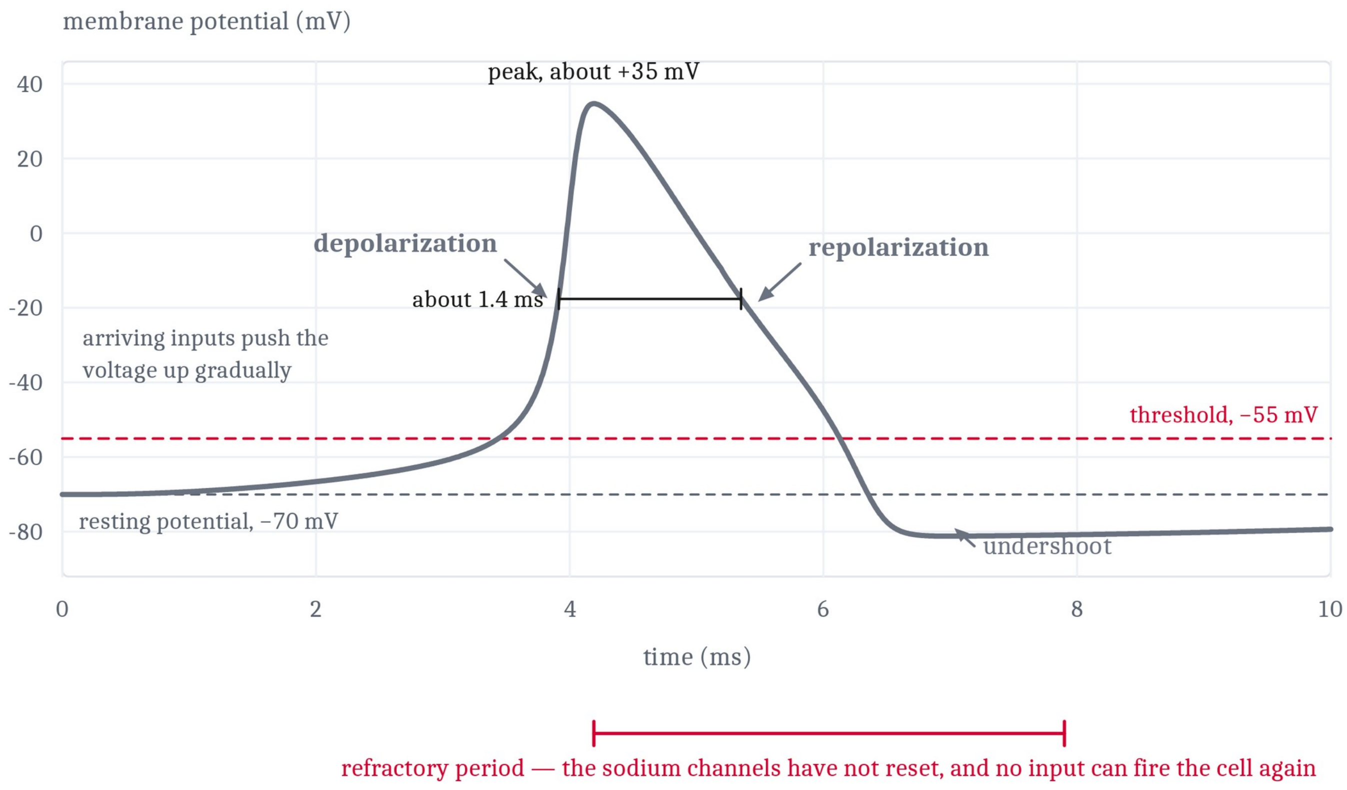
Sodium concentrated outside, potassium inside, both positive — and the inside negative anyway. That imbalance is the **resting membrane potential**, and the **sodium-potassium pump** holds it there, spending a molecule of **ATP** per cycle. **A neuron doing nothing is spending energy continuously, and what it buys is readiness.**



Ion counts are suggestive, not literal. A cell at rest is spending energy continuously, in order to stay this way.

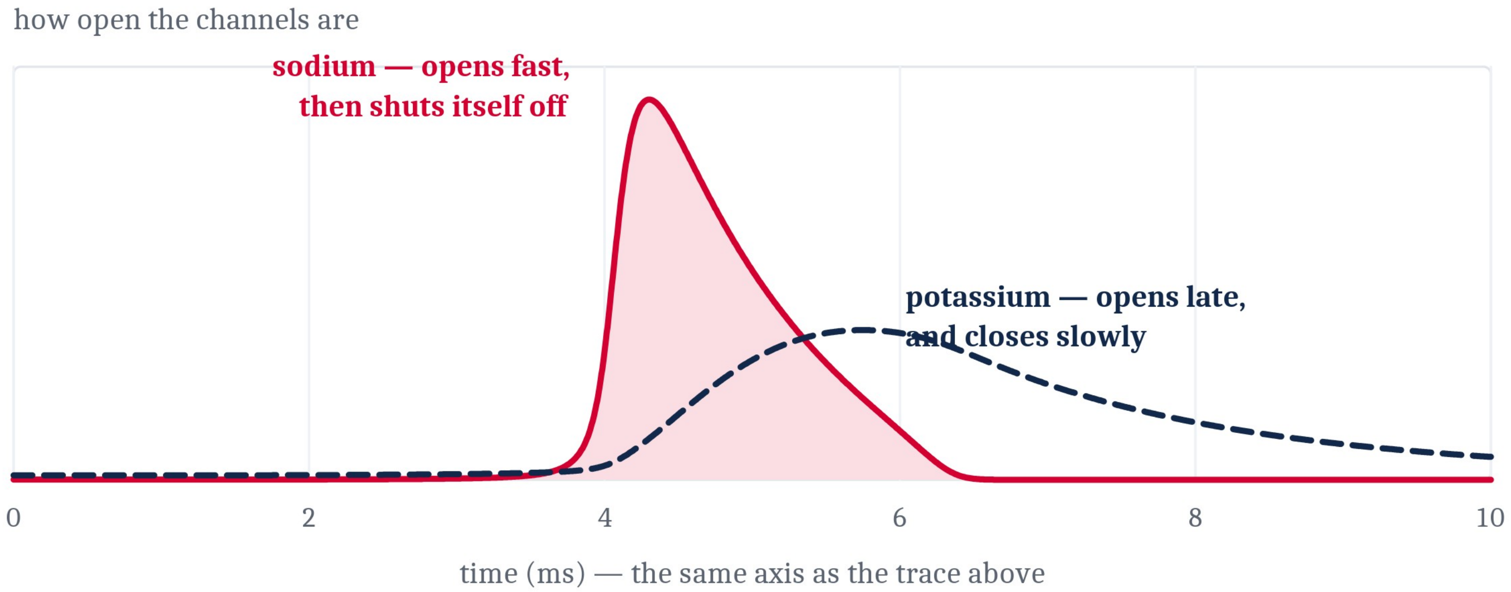
How a Spike Happens

Past a critical level, voltage-gated sodium channels open. Sodium rushes in, which pushes the voltage further up, which opens more channels. **Once that loop starts it cannot be stopped partway — depolarization**. Then the channels shut themselves off, potassium flows out — **repolarization** — and for a short interval no input can make the cell fire: the **refractory period**.



Why All-or-Nothing Was Never a Stipulation

A spike is all-or-nothing because the rising phase is a positive feedback loop, and a positive feedback loop has no intermediate settings. There is no way for a cell to produce half of an action potential — and the **refractory period** settles two more of the last lecture's assertions for free.



The runaway has no intermediate settings, which is why the spike is all-or-nothing.
And the cell cannot fire again until the sodium curve has come back down.

How Did Anyone Find This Out?

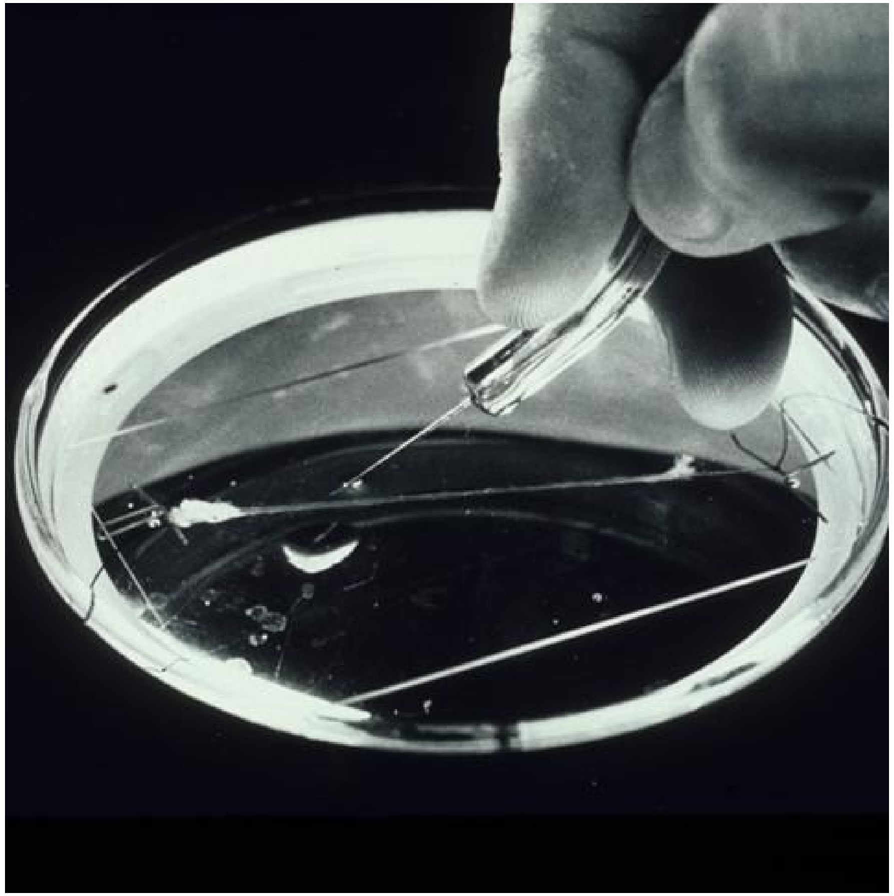
Everything so far is a story about invisible particles crossing a barrier we cannot watch. Stories like that are easy to tell and hard to believe.

Hodgkin and Huxley, late 1940s. The squid giant axon — a nerve fiber up to a millimeter thick, large enough to thread a wire down the inside of it.

They held the membrane voltage at fixed values, measured the currents, and wrote equations for how permeability to sodium and potassium changes with voltage and over time.

Then they calculated what an action potential should look like — and out came a curve with the right shape, height, duration, and conduction speed. None of it had been built in.

And nobody had ever seen an ion channel.



Two Models of One Neuron

Neither is a simplified version of the other, and neither is more true. They answer different questions, at different time scales, at different levels.

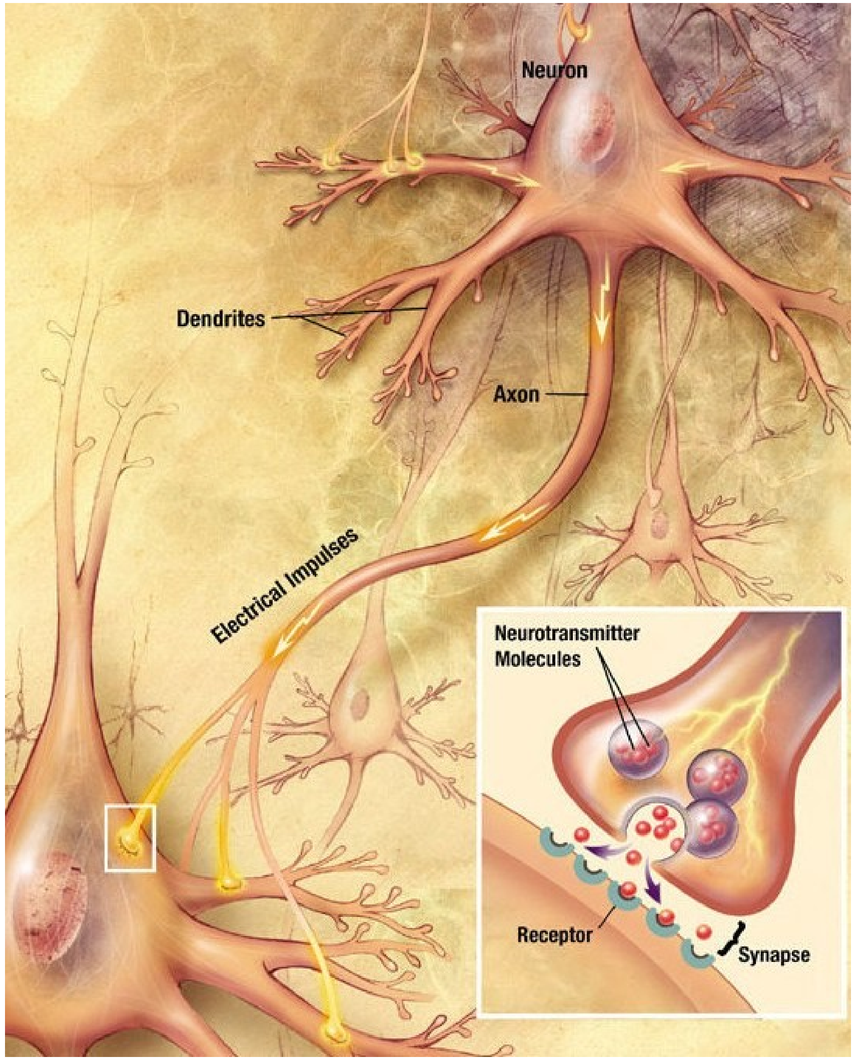
	The rate model	The Hodgkin-Huxley model
Marr's level	algorithmic	implementational
What it represents	firing rates, and a weight on each connection	currents crossing a membrane, and a voltage
The time scale it works at	long enough for behavior to happen	about a thousandth of a second
The question it answers	what does this cell contribute to a circuit?	how does a cell physically produce one event?
What it is built from	one line of arithmetic	equations fitted to measured currents
What it cannot say	how a spike is made	what an assembly of ten thousand cells does
Wire ten thousand together	the behavior of the whole can be worked out	enormous cost, and very little more learned

Down the Axon, and Across the Gap

A spike arrives at the far end **not as a single event traveling but as a chain of events regenerating** — each patch of membrane, as it depolarizes, pushes the patch ahead of it past threshold. Nothing weakens along the way.

Regenerating at every point is slow and expensive, so many vertebrate axons are wrapped in **myelin**, in segments with small bare gaps. The spike is regenerated only at the gaps and the stretches between are crossed passively — far faster, and considerably cheaper.

At the axon's end the arriving spike lets calcium in; calcium causes vesicles of neurotransmitter to fuse with the surface and spill into the gap. **Every stage of that costs energy** — ions pumped back, transmitters manufactured and loaded, vesicle membrane recovered and reused.



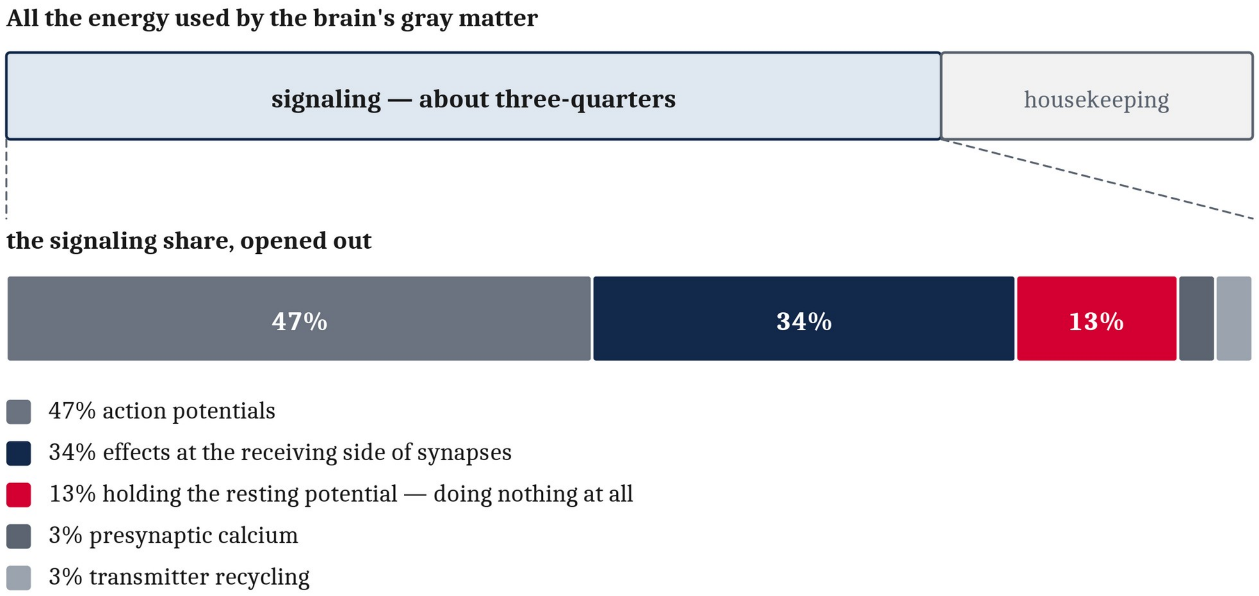
It Is the Channel, Not the Molecule

In the brain the dominant excitatory transmitter is **glutamate** and the dominant inhibitory one is **GABA**. But the difference between them is not really a fact about the two molecules. It is a fact about **the channels their receptors open.**

	Glutamate	GABA
Its receptors open channels permeable to	sodium	chloride
The charge that ion carries	positive	negative
Which way that charge moves	into the receiving cell	into the receiving cell
What the membrane voltage does	climbs toward threshold	is pushed down, away from threshold
The effect, named	excitatory	inhibitory

What Signaling Costs

Bio-energetics: how an organism gets, stores and spends energy. **The cost of computing with neurons is dominated not by the computing but by the cleanup.** Two findings, two different denominators — **three-quarters** of gray-matter energy goes to signaling rather than housekeeping, and almost all of *that* goes on the pump.



Almost all of the signaling share is spent by the sodium-potassium pump, restoring the gradients that signaling ran down.

A modeled budget at an assumed mean rate of 4 spikes per second, not a measurement. After Attwell & Laughlin (2001).

The Brain as a Metabolic Outlier

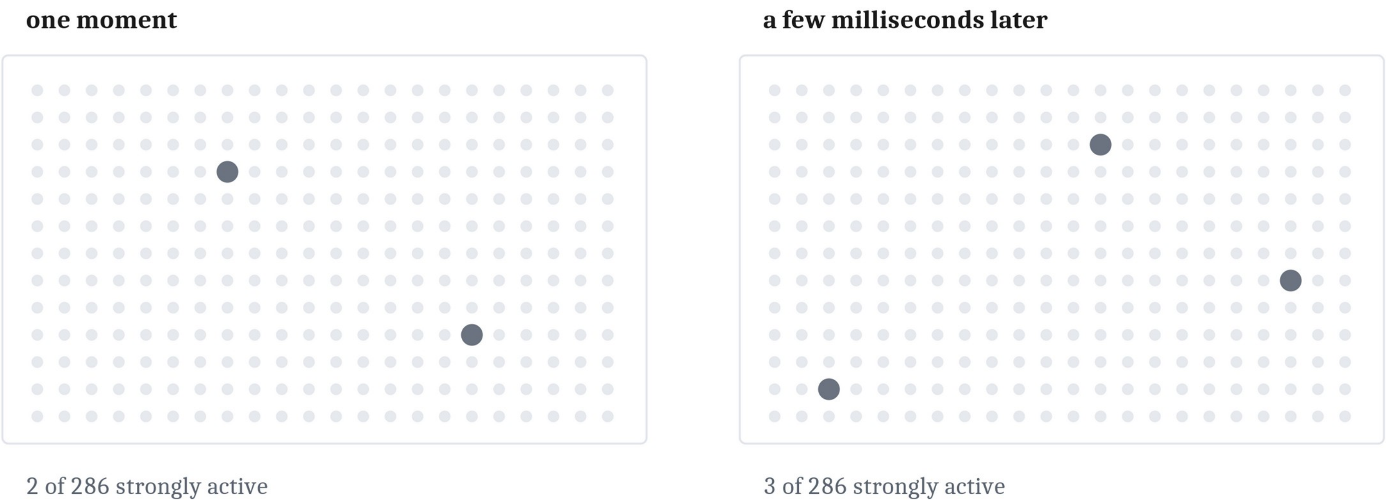
About two percent of body weight, about twenty percent of the body's oxygen at rest. Expressed as power rather than as a share, the same measurement is roughly the twenty watts from Module 0 — **one fact stated two ways.**

Measured against	The figure
the body, at rest	about 2% of body weight and about 20% of the oxygen used — one measurement stated two ways, and about twenty watts
a lifetime	as much as half of the body's oxygen use in the middle of the first decade of childhood, while the brain is still being built
other mammals	non-primate mammals spend 3–5% of the daily energy budget on the brain, other primates 8–10%, humans 20–25%
one neuron	about the same in a human as in any other primate

Energy as a Design Constraint

Run the accounting in reverse, and only a small proportion of cortical neurons can be strongly active at any one moment — on the order of **one in a few hundred**. That is **sparse coding**, and it is not an accident or an inefficiency.

The ten-percent claim is false in every sense that matters. What is true is stranger: which small share is firing changes moment to moment, so over any stretch of time the whole brain is in use. It is simply never all in use at once.



About one cortical neuron in a few hundred can be strongly active at any instant. That is what the energy budget affords.

Which few changes constantly, so over any stretch of time the whole brain is in use. It is simply never all in use at once.

The Same Function, and the Place It Strains

The last lecture's equation described a biological neuron and an artificial one about equally well. **Set the two implementations side by side and the resemblance vanishes completely — which is where the comparison between minds and machines actually strains.**

	A neuron	A transistor
What physically moves	ions, bodily, from one side of a membrane to the other	charge, along a conductor
Is chemistry involved	yes, and chemistry takes time	no
Time for one event	about a millisecond	a small fraction of a billionth of a second
What one event costs	ATP, spent hauling back every ion that crossed	electrical energy, most of it leaving as heat
Where the power comes from	food, and it has to be found every day	a supply outside the machine
What has to be carried away	waste heat, at a rate the body can manage	waste heat, and packing enough parts together makes it the limit
What those constraints force above	very many things at once, and most cells quiet	long sequences of steps, run very fast

Expensive and Cheap, and What Comes Next

We have built a neuron three times over — **once as a solution to a problem, once as a function, and once as a physical machine with a running cost.**

Expensive and cheap are both right

Expensive

measured against the animal's whole budget

20% everything else the body does

of the oxygen the body uses at rest —
spent by an organ that is about 2% of body weight.

One organ taking a fifth of everything.

A large share of a small budget.

Cheap

measured against the work done

20 watts

less than a dim light bulb

runs eighty-six billion neurons
and an entire human mind.

A small price for what it delivers.

The two verdicts do not conflict. They have different denominators.

Do not push the comparison with artificial systems too hard: the two are not doing the same job, and the accounting is not commensurable.

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